M3503 WEEK 6 2021

LAPLACE TRANSFORM

Let $f(t)$ be defined on $[0,\infty)$. The integral

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = F(s)$$

is the Laplace transform of $f(t)$ provided that it converges.

Examples

Determine the Laplace transform of

① $f(t) = 1$

② $f(t) = e^{at}$

③ $f(t) = \sin(at)$

④ $f(t) = \begin{cases} 0, & 0 < t < 2 \\ t, & t < 2 \end{cases}$

Soln

① $\mathcal{L}\{1\} = \int_0^\infty e^{-st}\,dt$

$$= \lim_{a\to\infty} \int_0^a e^{-st}\,dt$$

$$= \lim_{a\to\infty} \left[\frac{-1}{s} e^{-st}\right]_0^a$$

$$= \lim_{a\to\infty} \left[\frac{-1}{a} e^{-at} + \frac{1}{s}\right]$$

$$= \frac{1}{s}$$

② $\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st} e^{at}\, dt$

$= \lim_{N\to\infty} \int_0^N e^{(a-s)t}\, dt$

$= \lim_{N\to\infty} \left[\frac{1}{a-s} e^{(a-s)t}\right]_0^N$

$= \lim_{N\to\infty} \left[\frac{1}{a-s} e^{(a-s)N} - \frac{1}{a-s}\right]$

$= \frac{1}{s-a}$, if $a<s$

③ $\mathcal{L}\{\sin(at)\} = \int_0^\infty e^{-st} \sin(at)\, dt$

$= \lim_{N\to\infty} \int_0^N e^{-st} \sin(at)\, dt$

$= \lim_{N\to\infty} \left[\frac{s^2 e^{-st}}{s^2+a^2}\left(-\frac{\sin(at)}{s} - \frac{a\cos(at)}{s^2}\right)\right]_0^N$

$= \lim_{N\to\infty} \frac{s^2}{s^2+a^2}\left[-e^{-sN}\left(\frac{\sin(aN)}{s} + \frac{a}{s^2}\cos(aN)\right) + \frac{a}{s^2}\right]$

$= \frac{a}{s^2+a^2}$

④ $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$

$= \int_0^2 0\,dt + \int_2^\infty e^{-st} t\,dt$

$= \lim_{N\to\infty} \int_2^N e^{-st} t\,dt$

$= \lim_{N\to\infty} \left[-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right]_2^N$

$= \lim_{N\to\infty} \left[\frac{-N}{s} e^{-sN} - \frac{1}{s^2} e^{-sN} + \frac{2}{s} e^{-2s} + \frac{1}{s^2} e^{-2s} \right]$

$= \left(\frac{2}{s} + \frac{1}{s^2} \right) e^{-2s}$

Linearity Of Laplace Transform

Suppose Laplace transforms of f_1 and f_2 exist for $s > a$, then

① $\mathcal{L}\{f_1 + f_2\} = \mathcal{L}\{f_1\} + \mathcal{L}\{f_2\}$, for $s > a$

② $\mathcal{L}\{c f_1\} = c\,\mathcal{L}\{f_1\}$, for $s > a$

Exponential Order

A function $f(t)$ is said to be of exponential order c if there exist positive constants M and T such that

$|f(t)| \leq M e^{ct} \quad \forall\, t \geq T$

Piecewise Continuity

A function $f(t)$ is piecewise continuous on $[a,b]$ if it is continuous at every point in $[a,b]$, except for a finite number of points where $f(t)$ has a jump discontinuity.

Existence Of Laplace Trasform

If $f(t)$ is piecewise continuous on $[0,\infty)$ and is of exponential order a, then $\mathcal{L}\{f(t)\}=F(s)$ exists for $s>a$

Table of Laplace transforms

$f(t)$	$F(s)=\mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$
$t^n, n=1,2,\dots$	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(at)$	$\frac{a}{s^2+a^2}$
$\cos(at)$	$\frac{s}{s^2+a^2}$
$e^{at}t^n, n=1,2,\dots$	$\frac{n!}{(s-a)^{n+1}}$
$e^{at}\sin(kt)$	$\frac{k}{(s-a)^2+k^2}$
$e^{at}\cos(kt)$	$\frac{s-a}{(s-a)^2+k^2}$
$\sinh(kt)$	$\frac{k}{s^2-k^2}$
$\cosh(kt)$	$\frac{s}{s^2-k^2}$

PROPERTIES OF LAPLACE TRASFORM

① If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > \alpha$, then
$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$ for $s > \alpha + a$.
This is known as the first shift theorem.

② If $f(t)$ is continuous on $[0,\infty)$ and $f'(t)$ is piecewise continuous on $[0,\infty)$, with both of exponential order α, then

$$\mathcal{L}\{f'(t)\} = s\,\mathcal{L}\{f(t)\} - f(0), \text{ for } s > \alpha$$

③ If $f(t), f'(t), \ldots, f^{(n-1)}(t)$ are continuous on $[0,\infty)$ and $f^{(n)}(t)$ is piecewise continuous on $[0,\infty)$, with all these functions of exponential order α, then

$$\mathcal{L}\{f^{(n)}(t)\} = s^n \mathcal{L}\{f(t)\} - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots - f^{(n-1)}(0), \text{ for } s > \alpha$$

④ If $f(t)$ is piecewise continuous on $[0,\infty)$ and of exponential order α, then for $s > \alpha$

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}, \text{ where } F(s) = \mathcal{L}\{f(t)\}$$

⑤ $\lim_{s\to\infty} F(s) = 0$, where $F(s) = \mathcal{L}\{f(t)\}$

Examples

Find the Laplace transform of

① $f(t) = 2t^2e^{-t} - t + \cos(4t)$

② $f(t) = e^{-t}\, t \sin(2t)$

③ $f(t) = \sin^2(t)$

sol:

① $\mathcal{L}\{2t^2e^{-t} - t + \cos(4t)\}$

$= 2\mathcal{L}\{t^2e^{-t}\} - \mathcal{L}\{t\} + \mathcal{L}\{\cos(4t)\}$

$= 2\,\frac{2}{(s+1)^3} - \frac{1}{s^2} + \frac{s}{s^2+16}$

$= \frac{4}{(s+1)^3} - \frac{1}{s^2} + \frac{s}{s^2+16}$

② $\mathcal{L}\{e^{-t}\, t \sin(2t)\}$

$= -\frac{d}{ds}\mathcal{L}\{e^{-t}\sin(2t)\}$

$= -\frac{d}{ds}\left[\frac{2}{(s+1)^2+4}\right]$

$= -\frac{d}{ds}\left[\frac{2}{s^2+2s+5}\right]$

$= -2\left[\frac{-(2s+2)}{(s^2+2s+5)^2}\right]$

$= \frac{4(s+1)}{(s^2+2s+5)^2}$

(8) $\mathcal{L}\{\sin^2(t)\}$

$= \frac{1}{2}\mathcal{L}\{1 - \cos(2t)\}$

$= \frac{1}{2}\left[\mathcal{L}\{1\} - \mathcal{L}\{\cos(2t)\}\right.$

$= \frac{1}{2s} - \frac{s}{2(s^2+4)}$

Inverse Transform

Given a function $F(s)$, if there exists a function $f(t)$ that is continuous on $[0,\infty)$ and $\mathcal{L}\{f(t)\} = F(s)$, then $f(t)$ is the inverse transform of $F(s)$ i.e

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

Examples

Determine the inverse transform of

(1). $F(s) = \frac{1}{s^3}$

(2) $F(s) = \frac{6}{(s-1)^4}$

(3). $F(s) = \frac{s+1}{s^2+2s+10}$

(4). $F(s) = \frac{5s^2+34s+53}{(s+3)^2(s+1)}$

(5) $F(s) = \ln\left(\frac{s+2}{s-5}\right)$

① $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\}$

$= \frac{1}{2}t^2$

② $\mathcal{L}^{-1}\left\{\frac{6}{(s-1)^4}\right\} = \frac{6}{3!}\mathcal{L}^{-1}\left\{\frac{3!}{(s-1)^4}\right\}$

$= e^t t^3$

③ $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2s+10}\right\} = \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+9}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{s-(-1)}{(s-(-1))^2+3^2}\right\}$

$= e^{-t}\cos(3t)$

④ $\mathcal{L}^{-1}\left\{\frac{5s^2+34s+53}{(s+3)^2(s+1)}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{-1}{s+3} + \frac{2}{(s+3)^3} + \frac{6}{s+1}\right\}$ partial fractions

$= -e^{-3t} + 2te^{-3t} + 6e^{-t}$

⑤ $\mathcal{L}^{-1}\left\{\ln\left(\frac{s+2}{s-5}\right)\right\}$

$= -\frac{1}{t}\mathcal{L}^{-1}\left\{\frac{d}{ds}\ln\left(\frac{s+2}{s-5}\right)\right\}$

$= -\frac{1}{t}\mathcal{L}^{-1}\left\{\frac{-7}{(s+2)(s-5)}\right\}$

$= \frac{1}{t}\mathcal{L}^{-1}\left\{\frac{-1}{s+2} + \frac{1}{s-5}\right\}$

$= \frac{1}{t}\left[-e^{2t} + e^{5t}\right]$